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An Improved TOPSIS Method Using Fermatean Fuzzy Sets for Techno-Economic Evaluation of Multi-Type Power Sources

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Abstract

Scientific planning and optimal development of multi-type power sources are critical prerequisites for supporting the robust evolution of emerging power systems. However, existing techno-economic evaluation methods often face challenges such as higher-order uncertainty and weight conflicts, making it difficult to provide reliable support for comparing and selecting power source schemes. To address this, this paper proposes an improved Technique for Order Preference by Similarity to an Ideal Solution (TOPSIS) method based on Fermatean Fuzzy Sets (FFS) for techno-economic evaluation of multi-type power sources. First, building on the traditional TOPSIS framework, we introduce Fermatean Fuzzy Sets to construct a FF Hybrid Weighted Distance (FFHWD) measure. This measure simultaneously captures the subjective importance of evaluation indicators and decision-makers' risk preferences. Second, we design a subjective-objective coupled weighting strategy integrating Fuzzy Analytic Hierarchy Process (FAHP) and Entropy Weight Method (EWM) to achieve dynamic weight balancing, effectively mitigating biases caused by single weighting approaches. Finally, the FFHWD is integrated into the improved TOPSIS framework by defining FF positive and negative ideal solutions. The comprehensive closeness coefficients of each power source scheme are calculated to enable robust ranking and optimal selection of multi-type power source alternatives. Empirical analysis of five representative power generation technologies—thermal power, hydropower, wind power, photovoltaics (PV), and energy storage—demonstrates the following comprehensive techno-economic ranking: hydropower > photovoltaics > thermal power > wind power > energy storage. Hydropower achieves the highest closeness coefficient (−0.4198), whereas energy storage yields the lowest value (−2.8704), effectively illustrating their respective advantages and limitations within the evaluation framework. This research provides scientific decision-making support and methodological references for optimizing multi-type power source configurations and planning new power systems.

Keywords: multi-type power sources; techno-economic evaluation; Fermatean fuzzy; TOPSIS; hybrid weighting



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1. Introduction

Under the backdrop of accelerating energy structure transformation and building a new power system centered on renewable energy, the scientific evaluation and optimal

selection of multi-type power technologies have emerged as a critical priority for ensuring the security, stability, and low-carbon transition of power systems [1,2]. However, as wind and solar photovoltaic energy gain increasing shares in the energy mix, and energy storage systems develop into diverse flexible resources, power planning faces escalating challenges: heightened uncertainties on both supply and demand sides, entangled techno-economic indicators, and ambiguous long-term decision-making information [3,4]. Existing evaluation methods often fail to address high-order uncertainties and conflicting weight allocations, which undermines the reliability and precision of planning outcomes [5,6]. To address these limitations, this study proposes a novel framework to enhance decision-making robustness and rationality in scheme selection. Such advancements will not only support optimized power resource configurations but also advance coordinated planning of multi-type energy systems and the construction of next-generation power grids.

In the field of techno-economic evaluation of power generation technologies, researchers worldwide have explored various decision-making approaches to cope with different types of uncertainty. For example, Younis et al. introduced a probabilistic hesitant fuzzy set framework and developed a MARCOS-based multi-criteria decision-making method capable of integrating heterogeneous information for comparing technologies such as wind and solar power [7]. Building on the interplay between subjective knowledge and objective data, Chen et al. combined a modified FAHP with the entropy weight method to construct a hybrid weighting scheme, which they applied to evaluate the impacts of decommissioning power plants on system reliability and economic performance [8]. In another line of research, Li and Zhao proposed an enhanced fuzzy VIKOR approach for performance evaluation of eco-industrial power plants, improving both weight determination and aggregation mechanisms to better support decision-making under fuzzy environments [9]. Although these existing methods have proven effective in specific application scenarios, they often face limitations such as computational intensity, sensitivity to weight assignment, or insufficient ability to represent higher-order uncertainty. In contrast, the TOPSIS method remains attractive due to its transparent logic, efficiency, robustness, and intuitive interpretability. These advantages motivate the present study to adopt the TOPSIS framework while extending it with Fermatean fuzzy sets and a hybrid weighted distance measure, thereby addressing the challenges of uncertainty representation and weight conflict that persist in modern power system techno-economic evaluation.

Most existing studies on the techno-economic evaluation of power generation technologies are conducted under deterministic information conditions, lacking effective methods to address the inherent uncertainties in the evaluation process. Because the evaluation involves multidimensional and complex indicators, experts often find it difficult to describe them precisely using exact numerical values. As a result, the overall evaluation information tends to be vague and imprecise. Against this background, the FFS was proposed as an emerging fuzzy theory tool [10]. Compared with traditional Intuitionistic Fuzzy Sets (IFS) and Pythagorean Fuzzy Sets (PFS), FFS not only considers the degrees of membership and non-membership but also introduces a hesitation component. This allows it to represent a broader range of uncertainty information and provides greater adaptability and flexibility when dealing with complex evaluation problems [11,12]. Beyond these contributions, several studies have incorporated Fermatean fuzzy theory into evaluation frameworks to more effectively handle vagueness and hesitation in expert judgments. For instance, Gul et al. developed a Fermatean fuzzy TOPSIS model for industrial risk assessment, demonstrating its capability to quantify latent hazards [13]. Similarly, Yang and colleagues constructed a decision-making framework using a Fermatean fuzzy integrated weighted distance measure within the TOPSIS structure, enabling a more nuanced evaluation of green and low-carbon ports under high uncertainty [14].

Meanwhile, distance serves as an essential tool for measuring differences between data or variables, and it plays a pivotal role in multi-attribute decision-making methods [15,16]. For example, in the TOPSIS method, alternative schemes are ranked by calculating their distances from the ideal solution. Therefore, constructing an effective distance measurement approach has become a topic of significant research interest. Among various distance measures, the Ordered Weighted Distance (OWD) method allows flexible control over the influence of key data points by adjusting positional weights. This property has led to its widespread integration into numerous aggregation tools, giving rise to multiple extended forms. As a further development of OWD, the Hybrid Weighted Distance (HWD) method combines both ordered and arithmetic weighting mechanisms. It simultaneously accounts for the intrinsic importance of data and their positional influence during aggregation, thereby overcoming the limitations of the traditional OWD approach.

In this study, the HWD method is extended to the Fermatean fuzzy environment, and a new distance measure—FFHWD—is proposed to enhance the rationality and accuracy of the evaluation process. At the same time, to identify the relative importance of evaluation indicators, a comprehensive weighting model based on the FAHP and the EWM is constructed, integrating both expert subjective judgment and objective data information. Furthermore, the proposed FFHWD measure and the integrated weighting model are incorporated into the TOPSIS framework, forming a novel techno-economic evaluation model for multi-type power sources. This framework strategically integrates the TOPSIS methodology's inherent strengths—including structural transparency, computational efficiency, result robustness, and interpretability—while innovatively embedding the FFHWD measure and a hybrid weighting mechanism. This dual integration substantially enhances the framework's capacity to manage complex fuzzy information environments and establishes a more rigorous scientific foundation for weight determination through synergistic combination of subjective expert judgment and objective data analytics. An empirical analysis based on representative power sources in a Chinese province is conducted to verify the applicability of the model. The proposed framework provides a more scientific and robust decision-making basis for power project investment and operation management, thereby improving the practical effectiveness of engineering economic evaluation.

2. Evaluation Index System Based on DEMATEL

2.1. Initial Evaluation Indicator System

Constructing a techno-economic evaluation index system for multi-type power sources is the primary task of the assessment process, aiming to systematically and comprehensively reveal the core characteristics and attributes of the evaluated objects. The scientific soundness and rationality of the index system directly determine the accuracy and reliability of the evaluation results. Therefore, during its construction, several key principles must be strictly followed [17–20]. The selected indicators should be both scientific and practical, capable of objectively reflecting the techno-economic characteristics of power sources with clear objectives and broad applicability. They should also exhibit measurability and comparability, meaning that the indicator data can be easily obtained and allow for reliable horizontal and vertical comparisons. At the same time, the system should be comprehensive yet concise, covering all critical aspects of the techno-economic performance of power sources while maintaining a clear structure and concise formulation. Finally, the indicators should possess strong operability and independence to ensure ease of application and data processing, while minimizing semantic overlap or redundancy. Based on these principles, this study establishes the specific evaluation indicators for the techno-economic assessment of multi-type power sources as follows [21–23].

- (1) **Unit Electricity Cost.** This indicator reflects the total cost incurred for generating one kilowatt-hour of electricity over the entire life cycle of a power source. It is typically measured by the LCOE; for energy storage systems, the cycle-based cost per kilowatt-hour can be used instead. As the core metric for evaluating the economic competitiveness of power sources, the unit electricity cost directly determines their bidding capability in the electricity market and serves as the fundamental benchmark for economic comparison among different power types.
- (2) **Unit Capacity Investment Cost.** This indicator represents the initial investment required to construct each kilowatt of installed capacity. It reflects the project's financial threshold and investment pressure, thus serving as an important reference for investors and decision-makers when assessing project feasibility.
- (3) **Operation and Maintenance (O&M) Cost Ratio.** This refers to the proportion of annual O&M costs to the total annual generation revenue or overall cost. It indicates the operational stability and long-term economic performance of a power source. A high O&M cost ratio often implies a heavy operational burden and greater sensitivity of profitability to fluctuations in fuel prices or market electricity prices.
- (4) **Capacity Factor.** Defined as the ratio of actual electricity generation to the theoretical maximum generation under full-load operation, this indicator measures the “productivity” and utilization level of a power source. It reflects the combined effects of technological reliability and local resource conditions, thereby providing a comprehensive view of the power source's operational performance.
- (5) **Start-Up and Regulation Characteristics.** This composite performance indicator can be decomposed into two aspects—start-up time and ramp rate—and can also be evaluated systematically using Fermatean fuzzy numbers to quantify flexibility. In power systems with a high penetration of renewable energy, the rapid response and regulation capability of power sources play a crucial role in maintaining system stability and mitigating output fluctuations.
- (6) **Energy Conversion Efficiency.** This indicator represents the ratio of output energy to input energy. For power generation units, it corresponds to generation efficiency, while for energy storage systems, it reflects round-trip efficiency. It directly illustrates the technological advancement and energy utilization level of a system, where low efficiency typically indicates higher energy losses and hidden costs.
- (7) **Carbon Emission Intensity.** Carbon emission intensity measures the amount of CO₂-equivalent emissions produced per kilowatt-hour of electricity generated. Under the dual-carbon (carbon peaking and carbon neutrality) objectives, this indicator has become a key parameter for assessing the environmental friendliness of power sources. It directly affects their social acceptability, environmental costs, and long-term development prospects.

2.2. Dimensionality Reduction of the Evaluation Indicators Using DEMATEL

The DEMATEL method analyzes structural relationships within complex evaluation systems by calculating correlations among assessment indicators and classifying these indicators into cause factors and effect factors, thereby determining the importance levels of indicators within the evaluation system [24–26]. This study incorporates fuzzy mathematics into the traditional DEMATEL approach to address assessment uncertainty arising from semantic ambiguity. The correspondence between linguistic variables for expert judgments and fuzzy numbers is presented in Table 1.

Table 1. Correspondence between linguistic variables and triangular fuzzy numbers.

Scale	Degree of Influence	Triangular Fuzzy Number (l_{ij}, m_{ij}, u_{ij})
NO	No impact	(0, 0.1, 0.3)
VL	Slight impact	(0.1, 0.3, 0.5)
L	Minor impact	(0.3, 0.5, 0.7)
H	Significant impact	(0.5, 0.7, 0.9)
VH	Major impact	(0.7, 0.9, 1.0)

The steps for calculating centrality based on fuzzy DEMATEL are as follows:

- (1) Construct the initial direct influence matrix $\tilde{B}^{(k)}$. According to the correspondence between linguistic variables and fuzzy numbers in Table 1, the initial direct influence matrix $\tilde{B}^{(k)}$ provided by the k -th expert is obtained.
- (2) Calculate the normalized triangular fuzzy numbers ($ls_{ij}^{(k)}, ms_{ij}^{(k)}, us_{ij}^{(k)}$). Normalize the triangular fuzzy numbers in the initial direct influence matrix $\tilde{B}^{(k)}$ based on Equations (1)–(3).

$$ls_{ij}^{(k)} = \frac{l_{ij}^{(k)} - \min l_{ij}^{(k)}}{\Delta_{\min}^{\max}} \quad (1)$$

$$ms_{ij}^{(k)} = \frac{m_{ij}^{(k)} - \min m_{ij}^{(k)}}{\Delta_{\min}^{\max}} \quad (2)$$

$$us_{ij}^{(k)} = \frac{u_{ij}^{(k)} - \min u_{ij}^{(k)}}{\Delta_{\min}^{\max}} \quad (3)$$

where $\Delta_{\min}^{\max} = \max u_{ij}^{(k)} - \min l_{ij}^{(k)}$ represents the difference between the right-hand value and the left-hand value.

- (3) Calculate the comprehensive standardized value $x_{ij}^{(k)}$. First, calculate the left and right standard values $L_{ij}^{(k)}$ and $U_{ij}^{(k)}$ according to Equations (4) and (5). Then, calculate the comprehensive standardized value $x_{ij}^{(k)}$ based on Equation (6).

$$L_{ij}^{(k)} = \frac{ms_{ij}^{(k)}}{1 + ms_{ij}^{(k)} - ls_{ij}^{(k)}} \quad (4)$$

$$U_{ij}^{(k)} = \frac{us_{ij}^{(k)}}{1 + us_{ij}^{(k)} - ms_{ij}^{(k)}} \quad (5)$$

$$x_{ij}^{(k)} = \frac{L_{ij}^{(k)}(1 - L_{ij}^{(k)}) + U_{ij}^{(k)}U_{ij}^{(k)}}{1 - L_{ij}^{(k)} + U_{ij}^{(k)}} \quad (6)$$

- (4) Obtain the quantitative influence value $b_{ij}^{(k)}$ determined by the k -th expert for factor i on factor j .

$$b_{ij}^{(k)} = \min_{1 \leq k \leq K} l_{ij}^{(k)} + x_{ij}^{(k)} \Delta_{\min}^{\max} \quad (7)$$

- (5) Calculate the direct influence matrix B .

$$b_{ij} = (b_{ij}^{(1)} + b_{ij}^{(2)} + \dots + b_{ij}^{(k)}) / k \quad (8)$$

$$B = (b_{ij})_{n \times n} \quad (9)$$

- (6) Calculate the comprehensive influence matrix T by first standardizing the data in B according to Equation (10), and then obtaining the comprehensive influence matrix T based on Equation (11).

$$G = B / \max_{1 \leq i \leq n} \sum_{j=1}^n b_{ij} \quad (10)$$

$$T = G + G^2 + G^3 + \dots + G^n \quad (11)$$

- (7) Calculate the causality degree and centrality. Aggregate the elements in T separately by rows and columns, and calculate the influence degree m_i and the affected degree e_i of each evaluation indicator according to Equations (12) and (13). The causality degree e_i and centrality m_i can be calculated according to Equations (14) and (15).

$$f_i = \sum_{j=1}^n t_{ij}, \quad i = 1, 2, \dots, n \quad (12)$$

$$e_i = \sum_{i=1}^n t_{ij}, \quad j = 1, 2, \dots, n \quad (13)$$

$$n_i = f_i - e_i \quad (14)$$

$$m_i = f_i + e_i \quad (15)$$

If the causality degree $n_i > 0$, it can be determined that the indicator is a causal factor; otherwise, the factor is a result factor. The larger the centrality value m_i , the stronger the importance of the indicator in the entire evaluation system.

3. Combined Weighting Based on the Improved FAHP-EWM

3.1. Calculation of Subjective Weights Based on FAHP

In the techno-economic evaluation of multiple power generation technologies, experts' judgments on the relative importance of indicators often involve fuzziness and uncertainty. To address this, this study employs triangular fuzzy numbers to replace the precise values used in the traditional Analytic Hierarchy Process (AHP), thereby constructing a fuzzy judgment matrix. This approach effectively captures the fuzzy range and confidence level of expert evaluations. By applying the principle of fuzzy number comparison, it determines the ranking and weights of indicators, enhancing both the rationality and fault tolerance of the weighting process, and better reflecting real-world decision-making scenarios [27,28].

3.1.1. Construction of the Fuzzy Judgment Matrix

Experts employ the triangular fuzzy scale of FAHP to perform pairwise comparisons between each pair of criteria (i, j) , assessing the relative importance of criterion i over j [29–31]. The evaluation outcomes are then mapped to corresponding triangular fuzzy numbers $(l_{ijk}, m_{ijk}, u_{ijk})$. For each expert k , fuzzy judgment values for all (i, j) criterion pairs are obtained through the aforementioned method, resulting in the construction of a complete fuzzy judgment matrix A_k .

$$A_k = (\eta_{ijk})_{n \times n} = \begin{bmatrix} (l_{11k}, m_{11k}, u_{11k}) & (l_{12k}, m_{12k}, u_{12k}) & \dots & (l_{1nk}, m_{1nk}, u_{1nk}) \\ (l_{21k}, m_{21k}, u_{21k}) & (l_{22k}, m_{22k}, u_{22k}) & \dots & (l_{2nk}, m_{2nk}, u_{2nk}) \\ \vdots & \vdots & \ddots & \vdots \\ (l_{n1k}, m_{n1k}, u_{n1k}) & (l_{n2k}, m_{n2k}, u_{n2k}) & \dots & (l_{nnk}, m_{nnk}, u_{nnk}) \end{bmatrix} \quad (16)$$

where n is the number of indicators; l_{ijk} and u_{ijk} are the left and right judgment boundaries of the evaluation value, respectively, indicating the degree of fuzziness of the judgment. The larger they are, the higher the fuzziness of the judgment. m_{ijk} is the median value with a membership degree of 1, representing the most likely importance ratio judgment by experts, which is determined using the 1–9 scale method similar to the traditional AHP method. Among them, $\eta_{ijk} = (1, 1, 1)$, indicating that the indicator is equally important to itself. Moreover, when $\eta_{ijk} = (l_{ijk}, m_{ijk}, u_{ijk})$ is given, its inverse judgment must satisfy $\eta_{jik} = (1/u_{ijk}, 1/m_{ijk}, 1/l_{ijk})$, thus ensuring that the fuzzy judgment matrix meets the requirements of reciprocity and consistency.

3.1.2. Calculation of Weights at This Hierarchical Level

The fuzzy comprehensive importance of indicator i relative to the other indicators, denoted as Q_{ik} , is given as follows:

$$Q_{ik} = \left(\sum_{j=1}^n \eta_{ijk} \right) \otimes \left(\sum_{i=1}^n \sum_{j=1}^n \eta_{ijk} \right)^{-1} = \left(\frac{\sum_{j=1}^n l_{ijk}}{\sum_{i=1}^n \sum_{j=1}^n u_{ijk}}, \frac{\sum_{j=1}^n m_{ijk}}{\sum_{i=1}^n \sum_{j=1}^n m_{ijk}}, \frac{\sum_{j=1}^n u_{ijk}}{\sum_{i=1}^n \sum_{j=1}^n l_{ijk}} \right)^{-1} \quad (17)$$

$$\sum_{j=1}^n \eta_{ijk} = \left(\sum_{j=1}^n l_{ijk}, \sum_{j=1}^n m_{ijk}, \sum_{j=1}^n u_{ijk} \right) \quad (18)$$

$$\left(\sum_{i=1}^n \sum_{j=1}^n \eta_{ijk} \right)^{-1} = \left(\frac{1}{\sum_{i=1}^n \sum_{j=1}^n u_{ijk}}, \frac{1}{\sum_{i=1}^n \sum_{j=1}^n m_{ijk}}, \frac{1}{\sum_{i=1}^n \sum_{j=1}^n l_{ijk}} \right)^{-1} \quad (19)$$

where $\sum_{j=1}^n \eta_{ijk}$ represents the summation of triangular fuzzy numbers $\eta_{ijk} = (l_{ijk}, m_{ijk}, u_{ijk})$, and its summation rule follows the addition principle of triangular fuzzy numbers. When two triangular fuzzy numbers are $A = (l_1, m_1, u_1)$ and $B = (l_2, m_2, u_2)$, their sum is $A + B = (l_1 + l_2, m_1 + m_2, u_1 + u_2)$, because the left and right boundaries and the center value of triangular fuzzy numbers can be linearly superimposed independently, satisfying the closure and additivity of fuzzy number addition.

The degree of possibility that the comprehensive importance of indicator i , compared with the other indicators, is greater than that of indicator j ($j = 1, 2, \dots, n; j \neq i$), is denoted as $d(x_{ik})$ and is given as follows:

$$d(x_{ik}) = \min_{j=1,2,\dots,n;j \neq i} \left[\frac{l_{ijk} - u_{ijk}}{(m_{ijk} - u_{ijk}) - (m_{ijk} - l_{ijk})}, 1 \right] \quad (20)$$

The local fuzzy weight W_k of each indicator within the indicator set is given as follows:

$$W_k = \{d(x_{1k}), d(x_{2k}), \dots, d(x_{nk})\} \quad (21)$$

After normalization, the fuzzy weight set W'_k is obtained as follows:

$$W'_k = \{d'(x_{1k}), d'(x_{2k}), \dots, d'(x_{nk})\} \quad (22)$$

$$d'(x_{ik}) = \frac{d(x_{ik})}{\sum_{i=1}^n d(x_{ik})} \quad (23)$$

Based on the fuzzy weight set, the subjective weight W_s is calculated using the weighted arithmetic mean method as follows:

$$W_s = \lambda_k W'_k \quad (24)$$

where λ_k represents the weight of the k -th expert, $\sum_{k=1}^T \lambda_k = 1$.

3.2. Calculation of Objective Weights Based on EWM

To overcome the potential bias introduced by subjective weighting, this study employs the EWM to determine the objective weights of the evaluation indicators. This method automatically calculates the weights based on the degree of variation in the indicator data, thereby fully utilizing the information contained within the data itself. By minimizing human interference, it enhances the objectivity and interpretability of the techno-economic evaluation results for multiple power generation technologies [32].

3.2.1. Determination of System Entropy

Considering the safety condition of the j -th evaluation indicator as a system, the information entropy value e_j of this indicator can be derived according to the definition of entropy as follows:

$$e_j = -\frac{1}{\ln n} \sum_{i=1}^n y_{ij} \ln y_{ij} \quad (25)$$

$$y_{ij} = \frac{x'_{ij}}{\sum_{i=1}^n x'_{ij}} \quad (26)$$

3.2.2. Calculation of Weights

Based on the information entropy value of the j -th indicator, its weight w_j within this hierarchical level is calculated as follows:

$$\omega_j = \frac{1 - e_j}{\sum_{j=1}^n (1 - e_j)} \quad (27)$$

The information entropy value e_j may approach 1, in which case even a small change in entropy can lead to an excessively large deviation in the calculated weight. To address this issue, this study improves the weight calculation method. The modified weight ω'_j is computed as follows:

$$\omega'_j = \frac{\sum_{i=1}^n \frac{e_i}{n} + 1 - e_j}{\sum_{j=1}^n \left(\sum_{i=1}^n \frac{e_i}{n} + 1 - e_j \right)} \quad (28)$$

The subjective weights $W_s = (W_{s1}, W_{s2}, \dots, W_{sn})$ and the objective weights $W_o = (W_{o1}, W_{o2}, \dots, W_{on})$ are obtained using the FAHP and EWMs, respectively. The calculation formula for the comprehensive weights is as follows:

$$\omega_j = \delta W_s + (1 - \delta) W_o \quad (29)$$

where $\delta \in [0, 1]$.

4. Comprehensive Techno-Economic Evaluation Model for Multiple Power Generation Technologies Based on FFHWD-TOPSIS

4.1. Overview of the Traditional TOPSIS Method

In the techno-economic evaluation of multiple power generation technologies, the TOPSIS method ranks alternative schemes by quantifying their relative closeness to the ideal solution. This approach fully utilizes the information contained in the original data and clearly reveals the differences among alternatives. It offers a transparent computational process and produces stable and reliable results [33,34]. In this study, TOPSIS is incorporated into the evaluation framework because of its effectiveness in handling multi-attribute decision-making problems while integrating both subjective and objective weights. This provides a clear and interpretable basis for the optimal selection of power generation schemes. The modeling procedure is as follows:

Assume there are m alternative schemes, denoted as $A_1, A_2, \dots, A_m$, and n decision indicators, denoted as $R_1, R_2, \dots, R_n$. The decision matrix $X = (x_{ij})_{m \times n}$ constructed from the original data is given as follows:

$$X = \begin{bmatrix} \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ x_{m1} & x_{m2} & \cdots & x_{mn} \end{pmatrix} \end{bmatrix} \quad (30)$$

where x_{ij} represents the value of the j -th decision indicator for the i -th evaluation object.

- (1) Transform the original decision matrix X into the normalized decision matrix $Y = (y_{ij})_{m \times n}$ using the following formula. Normalizing the original decision matrix eliminates the influence of differing dimensions among indicators and resolves the issue of incomparability between them.

$$y_{ij} = \frac{x_{ij}}{\sqrt{\sum_{i=1}^n x_{ij}^2}} \quad (31)$$

- (2) Construct the weighted normalized decision matrix Z .

$$Z = (z_{ij})_{m \times n} \quad (32)$$

$$z_{ij} = W_j \times y_{ij} \quad (33)$$

where W_j represents the weight of the j -th evaluation indicator.

- (3) Determine the positive ideal solution S^+ and the negative ideal solution S^- . The positive ideal solution represents the scenario in which all evaluation indicators achieve their optimal values, while the negative ideal solution represents the scenario in which all indicators reach their worst values.

$$s_j^+ = \max\{z_{ij} / 1 \leq i \leq m\} \quad (34)$$

$$s_j^- = \min\{z_{ij} / 1 \leq i \leq m\} \quad (35)$$

- (4) Calculate the Euclidean distances of each alternative from the positive and negative ideal solutions:

$$d_i^+ = \sqrt{\sum_{j=1}^n (z_{ij} - s_j^+)^2} \quad (i = 1, 2, \dots, m) \quad (36)$$

$$d_i^- = \sqrt{\sum_{j=1}^n (z_{ij} - s_j^-)^2} \quad (i = 1, 2, \dots, m) \quad (37)$$

- (5) Calculate the relative closeness C_i of each evaluation object using the following formula. A larger C_i value indicates that the evaluation object is closer to the ideal solution, whereas a smaller C_i value indicates closer proximity to the negative ideal solution. The evaluation objects are then ranked according to the magnitude of their relative closeness values.

$$C_i = \frac{d_i^-}{d_i^+ + d_i^-} \quad (38)$$

Despite the well-structured framework and computational simplicity of the traditional TOPSIS method that have contributed to its widespread adoption in multi-attribute decision-making (MADM) domains, several critical limitations emerge, particularly in evaluation scenarios involving high uncertainty and strong fuzziness such as new energy power generation technologies. Firstly, the conventional TOPSIS assumes deterministic or precise numerical values for evaluation inputs, whereas practical techno-economic assessments of power generation systems often involve expert judgments characterized by fuzziness and hesitation. This discrepancy may result in information loss or bias during decision-making processes. Secondly, the default application of Euclidean distance to measure proximity between alternatives and ideal solutions fails to adequately account for variations in indicator importance and ranking sensitivity. The distance metric exhibits notable vulnerability to conflicting weight assignments, making evaluation outcomes excessively sensitive to extreme values or localized data fluctuations.

4.2. Fermatean Fuzzy Hybrid Weighted Distance

Given the limitations of traditional TOPSIS, it is essential to construct a distance metric with enhanced expressive capability and robustness to improve its adaptability in handling fuzzy, multi-scale evaluation information. This study introduces FFS to characterize membership, non-membership, and hesitation degrees in expert judgments [35–39], and further develops the FFHWD as a novel distance metric. This approach simultaneously integrates subjective-objective weights, positional weights, and fuzzy information while maintaining methodological simplicity. The proposed improvement preserves the structural integrity of traditional TOPSIS while significantly enhancing its stability and discriminative capability in complex techno-economic evaluations of power generation technologies.

First, let X be a non-empty set. The expression of an FFS Y belonging to X is given as follows:

$$Y = \{ \langle x_i, \mu_Y(x_i), \nu_Y(x_i) \rangle | x_i \in X \} \quad (39)$$

where $\mu_Y(x_i) : X \rightarrow [0, 1]$ is termed as ‘the membership degree of the factor x_i in the set Y ’, and $\nu_Y(x_i) : X \rightarrow [0, 1]$ is indicated as ‘the non-membership degree of the factor x_i in the set Y ’. In addition, $0 \leq [\mu_Y(x_i)]^3 + [\nu_Y(x_i)]^3 \leq 1$ for all $x_i \in X$. For a FFS Y and $x_i \in X$, $\tau = \sqrt[3]{1 - [\mu_Y(x_i)]^3 - [\nu_Y(x_i)]^3}$ is the indeterminacy degree of x_i to Y .

For simplicity, the FFN is denoted as $\alpha = \langle \mu_Y, \nu_Y \rangle$. Let λ be a positive real number, and let $\alpha = \langle \mu_Y, \nu_Y \rangle$, $\alpha_1 = \langle \mu_{Y1}, \nu_{Y1} \rangle$, and $\alpha_2 = \langle \mu_{Y2}, \nu_{Y2} \rangle$ be three FFNs defined on a non-empty set X . The corresponding operational definitions of these FFNs are given as follows:

$$\alpha_1 \oplus \alpha_2 = \left\langle \sqrt[3]{\mu_{Y1}^3 + \mu_{Y2}^3 - \mu_{Y1}^3 \mu_{Y2}^3}, \nu_{Y1} \nu_{Y2} \right\rangle \quad (40)$$

$$\alpha_1 \otimes \alpha_2 = \left\langle \mu_{Y1} \mu_{Y2}, \sqrt[3]{\nu_{Y1}^3 + \nu_{Y2}^3 - \nu_{Y1}^3 \nu_{Y2}^3} \right\rangle \quad (41)$$

$$\lambda \alpha = \left\langle \sqrt[3]{1 - (1 - \mu_Y^3)^\lambda}, \nu_Y^\lambda \right\rangle \quad (42)$$

$$\alpha^c = \langle \nu_Y, \mu_Y \rangle \quad (43)$$

For a FFN $\alpha = \langle \mu_Y, \nu_Y \rangle$, the functions $S(Y) = \mu_Y^3 - \nu_Y^3$ and $A(Y) = \mu_Y^3 + \nu_Y^3$ are defined as the score function and the accuracy function of α , respectively. If $S(\alpha_1) < S(\alpha_2)$, then $\alpha_1 < \alpha_2$; if $S(\alpha_1) = S(\alpha_2)$, then $\begin{cases} A(\alpha_1) < A(\alpha_2) \Rightarrow \alpha_1 < \alpha_2 \\ A(\alpha_1) = A(\alpha_2) \Rightarrow \alpha_1 = \alpha_2 \end{cases}$.

For two FFNs, $\alpha_1 = \langle \mu_{Y1}, \nu_{Y1} \rangle$ and $\alpha_2 = \langle \mu_{Y2}, \nu_{Y2} \rangle$, the distance between them can be defined as follows:

$$d(\alpha_1, \alpha_2) = \frac{1}{2} \left(\left| \mu_{Y1}^3 - \mu_{Y2}^3 \right| + \left| \nu_{Y1}^3 - \nu_{Y2}^3 \right| + \left| \tau_{Y1}^3 - \tau_{Y2}^3 \right| \right) \quad (44)$$

Let $A = (\alpha_1, \alpha_2, \dots, \alpha_n)$ be a set of FFNs, and let $w = (w_1, w_2, \dots, w_n)^T$ be the weight vector corresponding to these fuzzy numbers. Then, the Fermatean fuzzy weighted averaging (FFWA) operator is defined as follows:

$$FFWA(\alpha_1, \alpha_2, \dots, \alpha_n) = \sum_{j=1}^n w_j \alpha_j \quad (45)$$

To provide a more comprehensive characterization of both subjective and objective information in the evaluation of power supply schemes, this study adopts the concept of the HWD proposed in [40] and develops a FFHWD measure. This measure integrates both the intrinsic importance of each indicator and its positional weight within the sequence. By doing so, it more reasonably captures the decision-maker's risk preferences and the inherent differences within the data, thereby enhancing the overall comprehensiveness and reliability of the evaluation results [41,42].

For two sets of FFNS $A = (\alpha_1, \alpha_2, \dots, \alpha_n)$ and $B = (\alpha'_1, \alpha'_2, \dots, \alpha'_n)$, their FFHWD is defined as:

$$FFHWD_{w,\omega}(A, B) = \sum_{j=1}^n w_j \left(\tilde{d}(\alpha_{\sigma(j)}, \alpha'_{\sigma(j)}) \right) \quad (46)$$

where $\tilde{d}(\alpha_{\sigma(j)}, \alpha'_{\sigma(j)})$ denotes the j -th largest of the weighted individual distance $\tilde{d}(\alpha_j, \alpha'_j) = n\omega_j d(\alpha_j, \alpha'_j)$, $j = 1, 2, \dots, n$, $\omega = (\omega_1, \dots, \omega_n)^T$ is the weight vector related to the individual $d(\alpha_j, \alpha'_j)$, with $\omega_j \in [0, 1]$ and their sum is 1. $w = (w_1, \dots, w_n)^T$ is the weight vector for FFHWD measure. The balancing parameter n acts as a balance role.

4.3. Comprehensive Evaluation Framework Based on FFHWD-TOPSIS

Based on the previously established FFHWD measure and the combined weighting model, this section proposes a complete FFHWD-TOPSIS framework for the techno-economic evaluation of multiple power generation technologies. The proposed framework aims to systematically integrate subjective and objective information under a fuzzy envi-

ronment, providing a clear and operational procedure for scheme optimization. The main steps are illustrated in Figure 1.

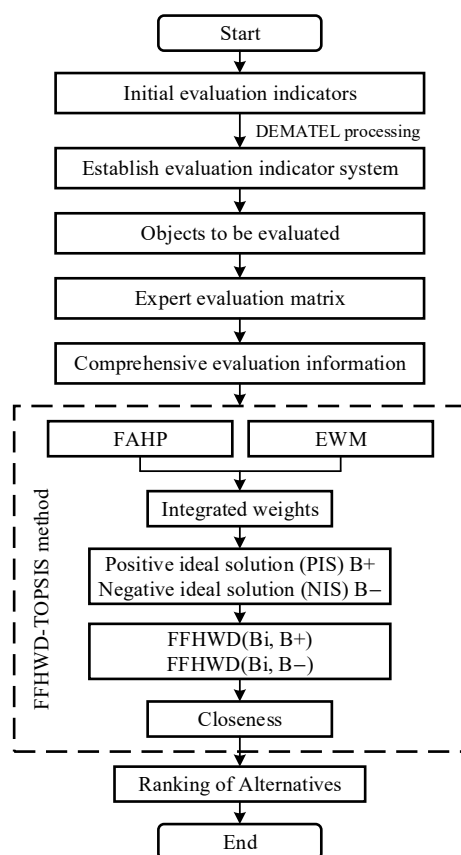


Figure 1. Flowchart of techno-economic evaluation for multi-type power sources.

- (1) Construct the Fermat-type decision matrix. Expert $e_k (k = 1, 2, \dots, K)$ evaluates the criterion $C_j (j = 1, 2, \dots, m)$ under the evaluation object $B_i (i = 1, 2, \dots, n)$ in the form of FFN, denoted as $\gamma^k = \langle \alpha_{ij}^k, \beta_{ij}^k \rangle$. Therefore, the fuzzy soft decision matrix $P^k = [\gamma_{ij}^k]_{n \times m}$ of the k -th expert can be obtained:

$$P^k = \begin{bmatrix} \langle \alpha_{11}^k, \beta_{11}^k \rangle & \langle \alpha_{12}^k, \beta_{12}^k \rangle & \cdots & \langle \alpha_{1m}^k, \beta_{1m}^k \rangle \\ \langle \alpha_{21}^k, \beta_{21}^k \rangle & \langle \alpha_{22}^k, \beta_{22}^k \rangle & \cdots & \langle \alpha_{2m}^k, \beta_{2m}^k \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle \alpha_{n1}^k, \beta_{n1}^k \rangle & \langle \alpha_{n2}^k, \beta_{n2}^k \rangle & \cdots & \langle \alpha_{nm}^k, \beta_{nm}^k \rangle \end{bmatrix}_{n \times m} \quad (47)$$

- (2) Normalize the individual decision matrices of experts. Let the normalized criterion value be $g_{ij} (i = 1, 2, \dots, n; j = 1, 2, \dots, m)$.
- (3) Apply the FFWA operator, combined with the weight of each expert $e_k (k = 1, 2, \dots, K)$ denoted as ε_k , to aggregate the decision matrices of all experts, thereby obtaining the overall fuzzy soft decision matrix $P = [\gamma_{ij}]_{n \times m} (i = 1, 2, \dots, n; j = 1, 2, \dots, m)$:

$$P = \begin{bmatrix} \langle \widetilde{\alpha}_{11}, \widetilde{\beta}_{11} \rangle & \langle \widetilde{\alpha}_{12}, \widetilde{\beta}_{12} \rangle & \cdots & \langle \widetilde{\alpha}_{1m}, \widetilde{\beta}_{1m} \rangle \\ \langle \widetilde{\alpha}_{21}, \widetilde{\beta}_{21} \rangle & \langle \widetilde{\alpha}_{22}, \widetilde{\beta}_{22} \rangle & \cdots & \langle \widetilde{\alpha}_{2m}, \widetilde{\beta}_{2m} \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle \widetilde{\alpha}_{nm}, \widetilde{\beta}_{nm} \rangle & \langle \widetilde{\alpha}_{nm}, \widetilde{\beta}_{nm} \rangle & \cdots & \langle \widetilde{\alpha}_{nm}, \widetilde{\beta}_{nm} \rangle \end{bmatrix}_{n \times m} \quad (48)$$

- (4) Apply the combined weighting method presented in Section 3 to determine the comprehensive weights of each indicator.
- (5) Calculate the Fermatean fuzzy PIS B^+ and Fermatean fuzzy NIS B^- as follows:

$$B^+ = \{C_1(B^+), C_2(B^+), \dots, C_n(B^+)\} = \left\{ \max_i \langle S(\gamma_{ij}) \rangle \mid j = 1, 2, \dots, n \right\} \quad (49)$$

$$B^- = \{C_1(B^-), C_2(B^-), \dots, C_n(B^-)\} = \left\{ \min_i \langle S(\gamma_{ij}) \rangle \mid j = 1, 2, \dots, n \right\} \quad (50)$$

- (6) Calculate the deviations between each alternative B_i and the FF PIS B^+ and NIS B^- , denoted as $FFHWD_{w,\omega}(B_i, B^+)$ and, respectively.
- (7) Calculate the closeness value $\zeta(B_i)$ for each alternative solution $B_i (i = 1, 2, \dots, n)$.

$$\zeta(B_i) = \frac{FFHWD_{w,\omega}(B_i, B^-)}{\max(FFHWD_{w,\omega}(B_i, B^-))} - \frac{FFHWD_{w,\omega}(B_i, B^+)}{\min(FFHWD_{w,\omega}(B_i, B^+))} \quad (51)$$

- (8) Sort all alternative schemes in descending order based on the closeness degree $\zeta(B_i)$ calculated in the previous step, and determine the optimal scheme.

5. Case Studies

5.1. Selection of Evaluation Indicators

To validate the effectiveness of the proposed techno-economic evaluation model, five representative power generation technologies were selected as case studies: thermal power (B_1), hydropower (B_2), wind power (B_3), photovoltaics (B_4), and energy storage (B_5). This combination encompasses traditional fossil fuels, renewable energy sources, and flexible resources, comprehensively reflecting the techno-economic characteristics and structural differences across multi-type power generation systems. The evaluation framework consists of seven core indicators: C_1 (levelized cost of electricity, LCOE), C_2 (capacity factor), C_3 (start-up and regulation characteristics), C_4 (energy conversion efficiency), C_5 (carbon emission intensity), C_6 (specific investment cost), and C_7 (operational and maintenance cost ratio), which systematically characterize the techno-economic performance of different power sources. Based on this framework, domain experts were invited to assess the interrelationships among the aforementioned indicators using the linguistic variables defined in Table 1. The DEMATEL method was then applied to structurally analyze the evaluation results. As an illustrative example, the data from Expert 1 are presented in Table 2.

Table 2. Expert 1's influence degree judgments for evaluation indicators.

Evaluation Indicator	C_1	C_2	C_3	C_4	C_5	C_6	C_7
C_1	NO	L	VH	H	VH	VH	H
C_2	H	NO	VH	H	H	L	VH
C_3	VH	H	NO	VH	VH	VH	H
C_4	L	VH	VH	NO	H	H	VH
C_5	VH	H	VH	L	NO	VH	H
C_6	VL	VL	L	H	VL	NO	L
C_7	L	VL	L	VL	VL	L	NO

Based on the correspondence between linguistic variables and fuzzy numbers in Table 1, and combined with the data in Table 2, the initial direct influence matrix provided by Expert 1 is as follows:

$$\tilde{B}^{(1)} = \begin{bmatrix} (0, 0.1, 0.3) & (0.3, 0.5, 0.7) & (0.7, 0.9, 1.0) & (0.5, 0.7, 0.9) & (0.7, 0.9, 1.0) & (0.7, 0.9, 1.0) & (0.5, 0.7, 0.9) \\ (0.5, 0.7, 0.9) & (0, 0.1, 0.3) & (0.7, 0.9, 1.0) & (0.5, 0.7, 0.9) & (0.5, 0.7, 0.9) & (0.3, 0.5, 0.7) & (0.7, 0.9, 1.0) \\ (0.7, 0.9, 1.0) & (0.5, 0.7, 0.9) & (0, 0.1, 0.3) & (0.7, 0.9, 1.0) & (0.7, 0.9, 1.0) & (0.7, 0.9, 1.0) & (0.5, 0.7, 0.9) \\ (0.3, 0.5, 0.7) & (0.7, 0.9, 1.0) & (0.7, 0.9, 1.0) & (0, 0.1, 0.3) & (0.5, 0.7, 0.9) & (0.5, 0.7, 0.9) & (0.7, 0.9, 1.0) \\ (0.7, 0.9, 1.0) & (0.5, 0.7, 0.9) & (0.7, 0.9, 1.0) & (0.3, 0.5, 0.7) & (0, 0.1, 0.3) & (0.7, 0.9, 1.0) & (0.5, 0.7, 0.9) \\ (0.1, 0.3, 0.5) & (0.1, 0.3, 0.5) & (0.3, 0.5, 0.7) & (0.5, 0.7, 0.9) & (0.1, 0.3, 0.5) & (0, 0.1, 0.3) & (0.3, 0.5, 0.7) \\ (0.3, 0.5, 0.7) & (0.1, 0.3, 0.5) & (0.3, 0.5, 0.7) & (0.1, 0.3, 0.5) & (0.1, 0.3, 0.5) & (0.3, 0.5, 0.7) & (0, 0.1, 0.3) \end{bmatrix}$$

Based on Equations (1)–(9), the initial direct influence matrix can be obtained as follows:

$$B^{(1)} = \begin{bmatrix} 0.1333 & 0.5 & 0.8667 & 0.7 & 0.8667 & 0.8667 & 0.7 \\ 0.7 & 0.1333 & 0.8667 & 0.7 & 0.7 & 0.5 & 0.8667 \\ 0.8667 & 0.7 & 0.1333 & 0.8667 & 0.8667 & 0.8667 & 0.7 \\ 0.5 & 0.8667 & 0.8667 & 0.1333 & 0.7 & 0.7 & 0.8667 \\ 0.8667 & 0.7 & 0.8667 & 0.5 & 0.1333 & 0.8667 & 0.7 \\ 0.3 & 0.3 & 0.5 & 0.7 & 0.3 & 0.1333 & 0.5 \\ 0.5 & 0.3 & 0.5 & 0.3 & 0.3 & 0.5 & 0.1333 \end{bmatrix}$$

Subsequently, by integrating the opinions of the remaining experts and applying the traditional DEMATEL method, the influence degree, affected degree, causality degree, and centrality of the evaluation indicators were calculated using Equations (10)–(15), with the detailed results presented in Table 3.

Table 3. Specific calculation results of evaluation indicators.

Evaluation Indicator	Nfluence Degree f_i	Affected Degree e_i	Causality Degree n_i	Centrality m_i	Normalized Centrality
C ₁	1.1108	0.7000	0.4108	1.8108	0.349
C ₂	1.0730	0.9644	0.1086	2.0374	0.733
C ₃	1.1276	1.0674	0.0602	2.1950	1.000
C ₄	1.1071	0.9194	0.1877	2.0265	0.715
C ₅	0.9156	0.8789	0.0367	1.7945	0.321
C ₆	0.6203	1.0601	−0.4398	1.6804	0.128
C ₇	0.6203	0.9845	−0.3642	1.6048	0.000

The centrality analysis in Table 3 indicates that evaluation indicator C₇ has the lowest influence on the overall system, followed by C₆. To simplify model complexity and focus on core elements, this study retains the five key indicators C₁–C₅ for subsequent evaluation. Based on this framework, four senior experts in power system planning (E1–E4) were invited to conduct evaluations using FFNs. Prior to the formal assessment, experts were provided with a Fermatean Fuzzy Number Operations Manual to define the semantic scales for the membership degree μ (“confidence in superior indicator performance”) and non-membership degree ν (“confidence in inferior indicator performance”). Experts were instructed to adhere to the constraint $\mu^3 + \nu^3 \leq 1$. A structured questionnaire was employed to independently assess FFNs for the five power generation technologies across the five indicators. The questionnaire utilized a 9-point linguistic variable system mapped to FFNs to ensure consistent interpretation among experts. The raw evaluation data are detailed in Table 4.

Table 4. Decision matrix of experts.

Expert	Alternative	C ₁	C ₂	C ₃	C ₄	C ₅
E1	B ₁	(0.61, 0.61)	(0.69, 0.34)	(0.66, 0.32)	(0.78, 0.16)	(0.62, 0.36)
	B ₂	(0.60, 0.41)	(0.65, 0.41)	(0.81, 0.51)	(0.57, 0.26)	(0.86, 0.82)
	B ₃	(0.47, 0.70)	(0.73, 0.56)	(0.71, 0.43)	(0.44, 0.24)	(0.66, 0.55)
	B ₄	(0.39, 0.47)	(0.72, 0.35)	(0.83, 0.32)	(0.65, 0.54)	(0.84, 0.66)
	B ₅	(0.78, 0.53)	(0.57, 0.67)	(0.75, 0.23)	(0.90, 0.72)	(0.55, 0.58)
E2	B ₁	(0.55, 0.90)	(0.71, 0.48)	(0.77, 0.07)	(0.78, 0.35)	(0.65, 0.33)
	B ₂	(0.63, 0.36)	(0.49, 0.82)	(0.67, 0.58)	(0.81, 0.23)	(0.39, 0.26)
	B ₃	(0.66, 0.55)	(0.58, 0.73)	(0.74, 0.64)	(0.76, 0.75)	(0.57, 0.52)
	B ₄	(0.84, 0.24)	(0.76, 0.64)	(0.87, 0.51)	(0.82, 0.64)	(0.67, 0.42)
	B ₅	(0.65, 0.37)	(0.60, 0.42)	(0.67, 0.23)	(0.81, 0.68)	(0.79, 0.33)
E3	B ₁	(0.64, 0.33)	(0.76, 0.42)	(0.66, 0.53)	(0.88, 0.37)	(0.70, 0.23)
	B ₂	(0.82, 0.21)	(0.86, 0.43)	(0.64, 0.23)	(0.79, 0.41)	(0.47, 0.77)
	B ₃	(0.67, 0.48)	(0.81, 0.47)	(0.59, 0.45)	(0.78, 0.44)	(0.39, 0.14)
	B ₄	(0.51, 0.20)	(0.66, 0.41)	(0.81, 0.54)	(0.60, 0.27)	(0.68, 0.55)
	B ₅	(0.71, 0.21)	(0.82, 0.43)	(0.77, 0.54)	(0.84, 0.32)	(0.84, 0.45)
E4	B ₁	(0.54, 0.22)	(0.46, 0.65)	(0.78, 0.53)	(0.58, 0.43)	(0.73, 0.43)
	B ₂	(0.57, 0.21)	(0.62, 0.40)	(0.85, 0.74)	(0.73, 0.57)	(0.64, 0.31)
	B ₃	(0.58, 0.65)	(0.69, 0.41)	(0.74, 0.31)	(0.38, 0.86)	(0.81, 0.38)
	B ₄	(0.80, 0.60)	(0.60, 0.15)	(0.57, 0.31)	(0.50, 0.65)	(0.70, 0.45)
	B ₅	(0.47, 0.34)	(0.58, 0.23)	(0.65, 0.52)	(0.67, 0.52)	(0.83, 0.32)

To mitigate potential sampling bias caused by the limited number of participating experts, this study applied 1000 bootstrap iterations to resample the expert evaluation matrices and calculate confidence intervals for the closeness coefficients of all alternatives. The results demonstrate that the rankings of all alternatives remain consistent within the 95% confidence interval, indicating the evaluation results exhibit strong robustness.

5.2. Model Application and Analysis

Based on the original expert evaluation data presented in Table 1, and considering their respective weights (0.3, 0.2, 0.25, 0.25), a collective decision matrix was obtained through weighted integration. Subsequently, the group evaluation results for the five power generation schemes were calculated, as shown in Table 5.

Table 5. Collective decision matrix.

	C_1	C_2	C_3	C_4	C_5
B_1	(0.5880, 0.5005)	(0.6540, 0.4655)	(0.7120, 0.3750)	(0.7550, 0.3180)	(0.6735, 0.3390)
B_2	(0.6535, 0.3000)	(0.6630, 0.4945)	(0.7495, 0.5115)	(0.7130, 0.3690)	(0.6135, 0.5680)
B_3	(0.5855, 0.6025)	(0.7100, 0.5340)	(0.6935, 0.4470)	(0.5740, 0.5470)	(0.6120, 0.3990)
B_4	(0.6125, 0.3890)	(0.6830, 0.3730)	(0.7680, 0.4105)	(0.6340, 0.5200)	(0.7310, 0.5320)
B_5	(0.6590, 0.3705)	(0.6410, 0.4500)	(0.7140, 0.3800)	(0.8095, 0.5620)	(0.7405, 0.4325)

Based on the group evaluation results in Table 5, the score function $S(Y)$ for each power generation scheme was first calculated to quantify their overall performance, as illustrated in Figure 2. On this basis, and according to Equations (39) and (40), the positive and negative ideal solutions B^+ and B^- under the Fermatean fuzzy environment were determined. The results are summarized in Table 6, providing the foundation for the subsequent distance calculation and scheme ranking.

**Figure 2.** Score functions of each power generation scheme.**Table 6.** Fermatean fuzzy positive ideal solution B^+ and negative ideal solution B^- .

	C_1	C_2	C_3	C_4	C_5
B^+	(0.6535, 0.3000)	(0.6830, 0.3730)	(0.7680, 0.4105)	(0.7550, 0.3180)	(0.7405, 0.4325)
B^-	(0.5855, 0.6025)	(0.6630, 0.4945)	(0.6935, 0.4470)	(0.5740, 0.5470)	(0.6135, 0.5680)

To scientifically assess the relative importance of each evaluation indicator in the comprehensive assessment process, the FAHP and EWMs were used to calculate the subjective and objective weights of each indicator, respectively. These were then combined using a fusion weighting approach with $\delta = 0.5$. The results are presented in Table 7. As shown in the table, indicator C_4 ranks first in both weighting methods, with a combined weight of 0.3145, indicating that its central role is reinforced by both data variation and expert consensus. Indicator C_3 follows with a weight of 0.242, highlighting the key importance of the capacity factor in the evaluation. In contrast, C_1 has the smallest weight of 0.087, suggesting a relatively limited influence. Overall, the integrated weighting results strike a

balance between subjective judgment and objective data, thereby enhancing the scientific rigor and credibility of the evaluation.

Table 7. Weights of index.

	C_1	C_2	C_3	C_4	C_5
FAHP	0.102	0.191	0.223	0.273	0.211
EWM	0.072	0.130	0.261	0.356	0.181
Integrated weights	0.087	0.1605	0.242	0.3145	0.196

To evaluate the overall performance of each power supply scheme, the FFHWD distances between each alternative and the Fermatean fuzzy positive and negative ideal solutions, $FFHWD(B_i, B^+)$ and $FFHWD(B_i, B^-)$, were first calculated. Based on these results, a weight vector was determined using an ordered weighted operator derived from the normal distribution. In this case, the vector was set as $(0.112, 0.236, 0.304, 0.236, 0.112)^T$. Subsequently, the relative closeness values $\zeta(B_i)$ were computed to measure how close each alternative is to the ideal solution. According to the closeness values presented in Table 8, the five power supply types can be ranked in terms of their techno-economic performance as follows: Hydropower (B_2) exhibits the highest closeness value (-0.4198), indicating the best overall performance. Photovoltaic power (B_4) and thermal power (B_1) rank second and third, respectively. In contrast, wind power (B_3) and energy storage (B_5) show relatively low closeness values (-1.8562 and -2.8704 , respectively). This suggests that both technologies are less competitive in the current evaluation framework—particularly energy storage, which performs poorly across several key indicators, leading to a lower overall ranking.

Table 8. Integrated weighted distance between alternatives with PIS and NIS.

	$FFHWD(B_i, B^+)$	$FFHWD(B_i, B^-)$	$\zeta(B_i)$	Ranking
B_1	0.0921	0.0979	-0.8736	3
B_2	0.0864	0.1521	-0.4198	1
B_3	0.1448	0.0829	-1.8562	4
B_4	0.0608	0.0767	-0.5003	2
B_5	0.2056	0.0827	-2.8704	5

Additionally, to systematically evaluate the impact of parameter δ , we conducted a sensitivity analysis. Table 9 presents the ranking outcomes of various power generation types and their corresponding relative closeness values when parameter δ is assigned different numerical values.

Table 9. Comparison of relative closeness and ranking orders under different parameter values.

δ	$\zeta(B_1)$	$\zeta(B_2)$	$\zeta(B_3)$	$\zeta(B_4)$	$\zeta(B_5)$	Ranking Order
$\delta = 0.1$	-0.7504	-0.4042	-1.7762	-0.5211	-2.7086	$B_2 > B_4 > B_1 > B_3 > B_5$
$\delta = 0.3$	-0.8078	-0.4148	-1.8304	-0.5176	-2.7853	$B_2 > B_4 > B_1 > B_3 > B_5$
$\delta = 0.5$	-0.8736	-0.4198	-1.8562	-0.5003	-2.8704	$B_2 > B_4 > B_1 > B_3 > B_5$
$\delta = 0.7$	-0.9386	-0.4257	-1.8869	-0.4762	-2.9437	$B_2 > B_4 > B_1 > B_3 > B_5$
$\delta = 0.9$	-1.0113	-0.4319	-1.9101	-0.4537	-3.0216	$B_2 > B_4 > B_1 > B_3 > B_5$

This sensitivity analysis demonstrates that the evaluation results exhibit high robustness when parameter δ varies within the range of 0.1 to 0.9: the ranking of the five power generation types remains consistently $B_2 > B_4 > B_1 > B_3 > B_5$ without any positional changes. As δ increases, the relative closeness values of all alternatives decrease overall, but their

relative gaps remain stable, confirming the method's low sensitivity to variations in the proportion of subjective and objective weighting. Notably, hydropower (B_2) consistently ranks first, while energy storage (B_5) shows a significant gap from the optimal solution (differing by over 2.0), highlighting its current technological and economic disadvantages.

5.3. Comparative Analysis for Model Validation

5.3.1. Comparison of Different Distance Measurement Methods

To verify the effectiveness and superiority of the proposed FFHWD method, a comparative analysis was conducted against two existing distance measures—FWAD and FOWD. Specifically, in the third step of the evaluation framework, the FWAD and FOWD measures were, respectively, applied to calculate the distances between each alternative and the Fermatean fuzzy positive (B^+) and negative (B^-) ideal solutions. Based on these calculations, two comparative evaluation frameworks were established, namely FWAD-TOPSIS and FOWD-TOPSIS. The corresponding relative closeness values of each evaluation object under the two frameworks were then obtained. The results are presented in Table 10 and Figure 3.

Table 10. Closeness $\zeta(B_i)$ of the alternative B_i .

	B_1	B_2	B_3	B_4	B_5
$\zeta_1(B_i)$	−0.4502	−0.1094	−1.1026	−0.5584	−1.9577
$\zeta_2(B_i)$	−1.3088	−0.4489	−1.8914	−0.1387	−3.4457

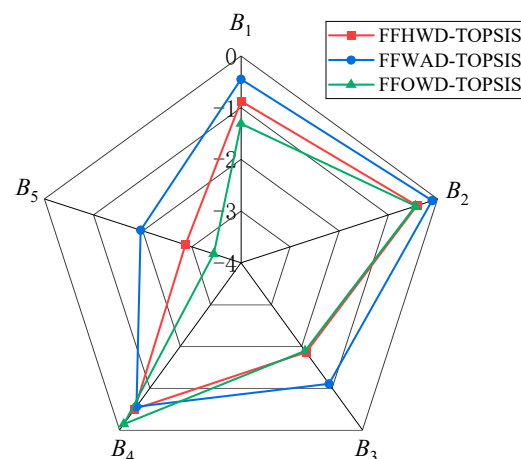


Figure 3. Relative closeness under different frameworks.

As shown by the ranking results above, the evaluation order obtained by FFWAD-TOPSIS is $B_2 > B_1 > B_4 > B_3 > B_5$, while that of FOWD-TOPSIS is $B_4 > B_2 > B_1 > B_3 > B_5$. The two methods identify B_2 and B_4 as the optimal alternatives, respectively, and exhibit noticeable differences in their overall rankings. The main reason lies in their weighting mechanisms. FFWAD focuses solely on the objective weights of different criteria and fails to incorporate the experts' subjective judgments. In contrast, FOWD captures the subjective preferences of decision-makers but overlooks the inherent importance differences among indicators. By comparison, the FFHWD method proposed in this study integrates the strengths of both bounded and arithmetic weighting schemes. It effectively balances subjective and objective information, thereby achieving more comprehensive data integration in the evaluation process and enhancing the rationality and stability of the final ranking results.

5.3.2. Comparison of Different Evaluation Methods

To thoroughly validate the validity and robustness of the proposed FFHWD-TOPSIS evaluation method, a systematic comparison is conducted against three widely adopted multi-criteria decision-making (MCDM) approaches in the field of energy system assessment: the VIKOR method based on the compromise solution principle, the MARCOS method integrating reference ideal solutions, and the classical hierarchical weighting AHP method. Table 11 summarizes the ranking outcomes of alternatives derived from these methods.

Table 11. Comparative results of different evaluation methods.

Evaluation Method	Ranking Result
VIKOR	$B_2 > B_1 > B_4 > B_3 > B_5$
MARCOS	$B_2 > B_4 > B_1 > B_3 > B_5$
AHP	$B_2 > B_1 > B_3 > B_4 > B_5$
FFHWD-TOPSIS	$B_2 > B_4 > B_1 > B_3 > B_5$

The comparative results show that all four methods consistently rank hydropower (B_2) as the top choice and energy storage (B_5) as the least preferred, validating the consensus in evaluation outcomes. The primary discrepancy lies in the ranking of thermal power (B_1) and photovoltaics (B_4): VIKOR and AHP prioritize thermal power over photovoltaics, whereas MARCOS and the proposed FFHWD-TOPSIS method yield superior rankings for photovoltaics. This discrepancy arises from the proposed method's use of Fermatean fuzzy sets to effectively capture higher-order uncertainties in evaluation information, combined with a hybrid weighted distance measure that simultaneously accounts for indicator importance and decision-makers' risk preferences. This dual mechanism enhances the scientific rigor of weight assignment and the rationality of distance calculations, thereby improving the robustness and interpretability of the ranking results. In contrast, traditional methods exhibit limitations in handling fuzziness and integrating weights, making them more prone to ranking biases.

Although the FFHWD-TOPSIS framework and the MARCOS method generate identical ranking outcomes in this specific numerical case, this consistency does not indicate equivalent modeling capabilities or decision robustness between the two approaches. The alignment in final rankings primarily results from the strong dominance relationships among the five alternatives across multiple critical criteria. These inherently stable ranking patterns can be captured by diverse multi-criteria decision-making techniques. More importantly, the methodological strengths of the FFHWD-TOPSIS approach extend beyond consistent results in a single dataset, manifesting more significantly in the following aspects:

- (1) Enhanced higher-order uncertainty handling capability. While the MARCOS method relies on standardized deterministic numerical inputs, FFHWD-TOPSIS explicitly incorporates Fermatean fuzzy membership degrees, non-membership degrees, and hesitation margins, enabling a more expressive representation of fuzzy expert knowledge.
- (2) Integrated hybrid weighting mechanism combining subjective and objective elements. The FFHWD measure simultaneously considers both inherent criterion importance and positional risk preferences, overcoming the purely data-driven limitations of MARCOS and effectively mitigating weight conflict sensitivity issues.
- (3) Superior ranking stability under parameter and data perturbations. As demonstrated in the sensitivity analysis, our approach maintains ranking robustness across variations in mixed weight proportions, whereas the MARCOS method lacks this flexible robustness modeling capability.

In summary, FFHWD-TOPSIS does not merely pursue numerical discrepancies with classical approaches on isolated datasets, but demonstrates superior generalization potential and expanded modeling capabilities in handling multi-source uncertainties, conflicting criteria integration, and disturbance resistance. The core objective of this comparative analysis is not to assert universal numerical superiority, but to validate that even under strong dominance structures, the proposed method can generate consistent and interpretable stable outcomes comparable to established MCDM techniques, while equipping evaluators with extended capabilities to address more complex fuzzy environments. This dual advantage constitutes its critical practical value in multi-type power generation assessment for next-generation power systems.

6. Conclusions

This study introduces a novel TOPSIS-based framework (FFHWD-TOPSIS) that integrates Fermatean fuzzy sets with a hybrid weighted distance measure and a combined FAHP–EWM weighting strategy, addressing higher-order uncertainty and weight conflicts in techno-economic evaluation. The FFHWD measure captures both subjective importance and decision-maker risk preferences and is integrated into an improved TOPSIS model via Fermatean fuzzy positive and negative ideal solutions.

- (1) Performance ranking: The FFHWD-TOPSIS model was applied to five representative power sources (thermal, hydropower, wind, photovoltaic, and energy storage) using expert evaluations. The computed closeness coefficients rank hydropower (B_2) highest in overall performance (best techno-economic score), with photovoltaic (B_4) and thermal power (B_1) in second and third place, respectively. In contrast, wind power (B_3) and energy storage (B_5) have much lower closeness values and thus appear less competitive under the current evaluation framework.
- (2) Comparative analysis: Compared to existing Fermatean-fuzzy TOPSIS variants using FWAD or FOWD distance measures, the proposed FFHWD-TOPSIS yields more balanced and stable rankings. By effectively blending objective and subjective weighting information, the method enhances the rationality and consistency of the final ranking.

These findings demonstrate that the FFHWD-TOPSIS framework provides reliable, nuanced decision support for optimizing multi-type power source configurations under uncertainty.

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